



International R&D partnerships and intrafirm R&D–marketing–production integration of manufacturing firms in emerging economies



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ABSTRACT

Although cross-functional integration is important for research and development (R&D), research about implications of cross-functional integration has been rather sparse. In new product development (NPD), no study to date has examined intrafirm as well as interfirm integration of key functions such as intrafirm R&D–marketing–production together with interfirm integration of host R&D–partner R&D. Such marketing and operations interface contributes to a better understanding of how operational and marketing activities impact on competitiveness and firm performance. This study collected data from 202 electronics manufacturing firms operating in an emerging economy, mainland China and Hong Kong with international R&D partnerships. The findings indicate that a high level of R&D integration between firms improved NPD performance when cross-functional integration is based on existing rather than new product configurations and key technologies. Interestingly, in high distance situations, cross-functional integration in the production validation stage generated NPD success. The findings show that high environmental uncertainties lead to a high level of host and partner firms R&D integration. However, product newness has no significant effects on R&D integration in any of the NPD stages.

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1. Introduction

In today's globalized markets, one of the ways firms respond to competitive pressures is by developing international research and development (R&D) partnerships, and strengthening cross-functional integration (Song, Thieme, & Xie, 1998; Van Dierdonck & Miller, 1980). In particular, firms in emerging markets increasingly form R&D partnerships with foreign firms to compete with established global firms and gain new knowledge such as new technologies and digitized product development processes. With rapid proliferation of new product offerings, fast changing environments and shortened product life cycles, knowledge of how integration of key functions and stages of new product development (NPD) in intra and interfirm integration affect successful operations will determine a firm's long-term competitiveness (Holland, Gaston, & Gomes, 2000; Koufteros, Vonderembse, & Doll, 2002; Verona, 1999). Operations research shows the importance of cross-functional integration among organizational functions in determining new product performance (e.g., Ernst, Hoyer, & Rübsaamen, 2010; Harryson, 1997). As global competition intensifies, it is imperative for firms operating in emerging economies (China, Brazil and India) to improve operational efficiency of functional interdependence in intra and interfirm R&D partnerships. Cross-functional integration can help firms

not only generate innovation but also reduce inefficiency of information asymmetry as a result of resource and/or activity duplications among functions and between firms. Yet, no research has examined the impact of both intra and interfirm integration activities on NPD performance.

Previous studies mainly examined intra-firm interaction and collaboration among functions, e.g., marketing, logistics, R&D, finance and manufacturing (e.g., Joshi, 2010; Kahn & Mentzer, 1994; Maltz & Kohli, 2000). However, it is important to examine intrafirm integration across functions such as R&D–marketing–production together with interfirm integration of host R&D–partner R&D' (hereafter R&D–R&D') because interfirm NPD collaboration can be affected by intrafirm cross-functional integration (e.g., marketing–manufacturing, R&D–marketing). Consideration of both intra and interfirm integration can provide new insights into functional interdependence and new product performance success from operational as well as industrial marketing perspectives. For instance, operational demands of cross-functional activities combined with marketing's emphases such as environmental situations or situational dimensions would provide a more complete picture than separate treatment of either field of study or intra and interfirm functional integration. In NPD literature, situational dimensions include product newness, physical distance, R&D experience and environmental uncertainty (Griffin & Hauser, 1996; Jin, 2001; Lu & Yang, 2004; Olson, Walker, Ruekert, & Bonner, 2001; Song & Parry, 1997; Song et al., 1998). Since differing situational dimensions would have different degrees of impact on different types of cross-functional integration and NPD stages, it is important to understand appropriate levels of functional integration especially for an interfirm NPD collaboration spanning diverse geographical

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